Assignment: Programming Basics

Type: Group Assignment

Important Rules

- Students should **ONLY USE** programming topics (e.g., variables, constants, arithmetic expressions, input/output statements, formatting, alignment, and decision-making statements) covered in unit 2-1 and unit 2-2 slides because the assignment assesses these concepts only.
- **The instructor reserves the right to ask questions about concepts, formats, or statements that are not covered in Unit 1, Unit 2-1, or Unit 2-2, and to apply grading penalties.**
- **Late submission will not be accepted**

Oil/Gas Station Scenario

A gas station wants to write a program which:

- Displays the following menu:
  - G: Gas
  - O: Oil
- Allows the user to select one of the above options. The user can only type (g or G) and (o or O).
  - Using lower() or upper() is permitted.
- If the user selects invalid option (not g/G or o/O), the program should display 'Invalid input.
- Calculates the cost of their oil and gas purchases from their wholesaler, including any discounts and GST. Calculation is based on the following business rules:
  - Oil must be purchased by the case. There are 12 litres/case.
    - Oil costs the station $1.25/ litre before discounts.

© 2026, Southern Alberta Institute of Technology

- If the station purchases more than 6 cases, they receive a 10% discount off the per litre price of the purchase.

- Gasoline is purchased by the litre at a cost of $1.05 per litre.
  - If the station purchases more than 2000 litres, they receive a 10% discount off the per litre price.

- Calculate GST according to the following table:

Province	GST %
AB or BC	5
ON	13
Others	15

  - The user can type the province in lower- or upper-case letters.

Equipment and Materials

For this assignment, you will need:

- Python IDE

Instructions

This assignment consists of three sections, all completed outside of class time. See the Brightspace for exact due date.

1. Review the Oil/Gas station scenario and the provided screenshots in the program testing section.
2. Design a flowchart to represent the process/algorithm of calculating the purchase summary (**No submission**).
3. Write a program that meets the requirements described in the above scenario and calculates the cost of oil or gas purchases.

   Your program should ask the user to enter the following information:

   - Number of oil cases or gas litres
   - Province
4. The program **MUST use constants** for number of litres per case (i.e., 12), oil and gas price (i.e., 1.25 and 1.05), and number of cases/litres (i.e., 6, 2000) which determines if the discount is applicable.

© 2026, Southern Alberta Institute of Technology

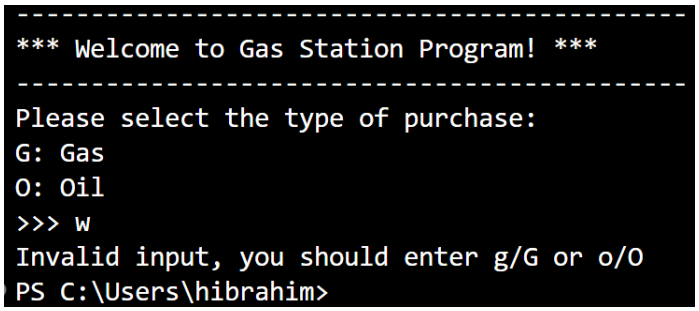
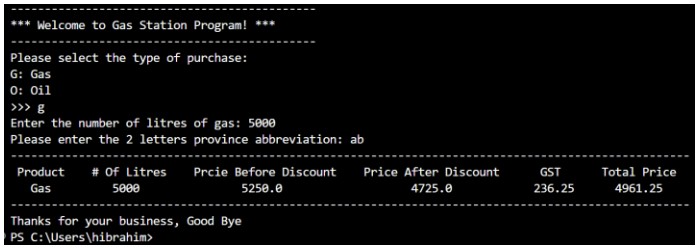
5. The program **must validate** number of oil cases and gas litres (the values must be greater than 0). If a value is invalid, the program should exit (hint: use exit())
6. When your program is complete, use the data listed in each of the following tests below to see if your program works correctly.

   Note: Check your program against the marking criteria for the submission.
7. Your program's output **MUST EXACTLY MATCH** the following tests.

Submission

- Submit the following to Brightspace:
  - The code of the program that you implemented (.py file)
  - A copy of the output from your test runs (.txt file)
  - The PDF/DOC peer assessment according to instructor's instructions.

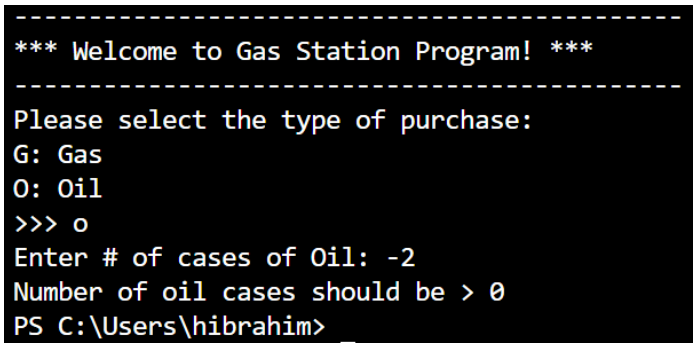
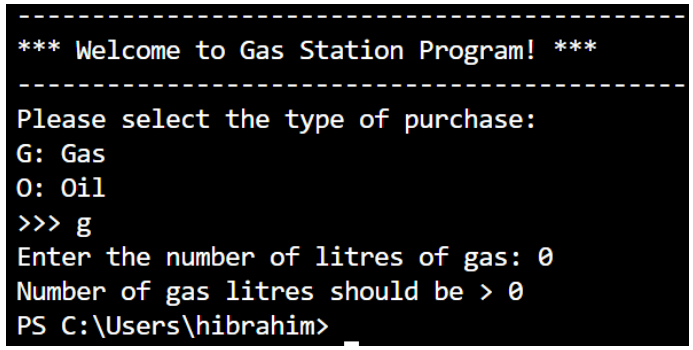
Program Testing

Test 1: User enters invalid option	
Test 2: User enters g/G, 5000 litres, and ab	

© 2026, Southern Alberta Institute of Technology

Test 3: User enters ‘g/G’, 1500 litres, and ‘on’	
Test 4: User enters ‘o/O’, 10 cases, and ‘ab’	
Test 5: User enters ‘o/O’, 5 cases, and ‘on’	
Test 6: User enters ‘o/O’, 6 cases, and ‘NL’	

© 2026, Southern Alberta Institute of Technology

Test 7: Invalid number of cases/litres	 

Peer assessment

- Only one DOC/PDF should be used for peer assessment.
- Final grades will be assigned according to members' participation.
- The peer assessment should be completed accurately and honestly
    - Deduction will be applied if it is not
- Each member should assess the contribution/participation of all other group members.
- Each member should assign a mark out of 10 to other members.
- Accurately describe the task completed by each member
    - Deduction will be applied if it is not

© 2026, Southern Alberta Institute of Technology

- Please use the following table to complete the peer assessment. Marks/tasks are just samples
    - Deduction will be applied if it is not used

	Member 1 Name	Member 2 Name	Member 3 Name	Completed Tasks
Member 1 Name	N/A	9	10	• Task 1 description • Task 2 description
Member 2 Name	8	N/A	9	• Task 3 description • Task 4 description
Member 3 Name	9	10	N/A	• Task 5 description • Task 6 description
Average	8.5	9.5	9.5	•

Marking Criteria

Check marking rubrics in BrightSpace

© 2026, Southern Alberta Institute of Technology